تربية وتعليم

مدرسة الخالدية الحلقة الثالثة بنين الشرقية

AL KHALIDIA BOYS' SCHOOL · C3 EASTERN

الميثاق الوطني للتعليم

END OF TERM 1 · ACADEMIC YEAR 2026 – 2027

Grade 10 Advanced

Exam Coverage Compilation

تجميعة هيكل امتحان نهاية الفصل الدراسي الأول

Reveal Integrated Math II | 25 questions | MCQ 1–20 · FRQ 21–25 | 150 min · /100

Draft coverage — for teachers' feedback only · not for students

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مدير المدرسة د/ عمر سليمان العضب

بالميثاق نبلغ الآفاق

Grade 10 Advanced · Coverage at a glance

Q	LESSON	LEARNING OUTCOME	EXAMPLE / EXERCISE	PAGE
1	**Lesson 11-1** Adding and Subtracting Polynomials	Write polynomials in standard form. Add and subtract polynomials	Exercises 10-14, 22-26	**P635**
2	**Lesson 11-3** Multiplying Polynomials	Multiply binomials by using the FOIL method	Exercises 1-6	**P649**
3	**Lesson 11-6** Factoring Quadratic Trinomials	Factor trinomials of the form x^2+bx+c	Exercises 1-6	**P671**
4	**Lesson 11-7** Factoring Special Products	Factor perfect square trinomials	Exercises 18-21	**P679**
5	**Lesson 2-1** Angles of Polygons	Find and use the sum of the measures of the interior angles of a polygon	Exercises 1-4	**P63**
6	**Lesson 2-2** Parallelograms	Recognize and apply the properties of the sides and angles of parallelograms	Exercises 9-12	**P72**
7	**Lesson 2-3** Tests for Parallelograms	Recognize the conditions that ensure a quadrilateral is a parallelogram	Learn · Exercises 1-6	**P75 · P79**
8	**Lesson 2-6** Trapezoids and Kites	Recognize and apply the properties of kites	Exercises 13-16	**P106**
9	**Lesson 3-1** Dilations	Find the scale factor of a dilation	Exercises 13-16, 20	**P120–121**

Grade 10 Advanced · Coverage at a glance

Q	LESSON	LEARNING OUTCOME	EXAMPLE / EXERCISE	PAGE
10	**Lesson 3-2** Similar Polygons	Solve problems using the properties of similar polygons	Exercises 11-14	**P128**
11	**Lesson 3-1** Dilations	Find the scale factor of a dilation	Exercises 10-12, 17-18	**P120**
12	**Lesson 3-3** Similar Triangles: AA Similarity	Identify similar triangles using the AA Similarity Postulate	Exercises 1-8	**P133**
13	**Lesson 3-5** Triangle Proportionality	Use proportional parts within triangles	Exercises 11-16	**P146**
14	**Lesson 3-6** Parts of Similar Triangles	Recognize and use proportional relationships of corresponding angle bisectors, altitudes, and medians of similar triangles	Exercises 1-11	**P153**
15	**Lesson 4-1** Geometric Mean	Find the geometric mean between two numbers	Exercises 1-6, 11-14	**P165**
16	**Lesson 4-2** Pythagorean Theorem and Its Converse	Use the Pythagorean Theorem and its converse	Exercises 7-12 · 20-22	**P171 · P172**
17	**Lesson 4-4** Special Right Triangles	Use the properties of 45°-45°-90° triangles and use the properties of 30°-60°-90° triangles	Exercises 19-25, 33-38	**P184–185**
18	**Lesson 4-5** Trigonometry	Use trigonometric ratios to find side lengths and angle measures of right triangles	Exercises 22-27, 30-32	**P192–193**

Grade 10 Advanced · Coverage at a glance

Q	LESSON	LEARNING OUTCOME	EXAMPLE / EXERCISE	PAGE
19	**Lesson 4-6** Applying Trigonometry	Use trigonometric ratios to find side lengths and angle measures of right triangles	Exercises 1-7	**P199**
20	**Lesson 4-6** Applying Trigonometry	Find the area of a triangle using two sides and an included angle	Exercises 13-21	**P200–201**
21	**Lesson 11-7** Factoring Special Products	Factorize binomials that are the difference of squares and factor perfect square trinomials	Exercises 35-41, 51-56	**P680–681**
22	**Lesson 2-1** Angles of Polygons	Find and use the sum of the measures of the exterior angles of a polygon	Exercises 9-18	**P64**
23	**Lesson 3-4** Similar Triangles: SSS and SAS Similarity	Use similar triangles to solve problems	Exercises 5-11	**P139**
24	**Lesson 4-3** Coordinates in Space	Find the distance between two points on the coordinate plane	Exercises 13-18 · 19-24	**P177 · P178**
25	**Lesson 4-7** The Law of Sines	Determine whether three given measures of a triangle define 0, 1, or 2 triangles by using the Law of Sines	Exercises 15-26	**P208**

◆ LESSON 11-1 · Adding and Subtracting Polynomials

Example 2

Write each polynomial in standard form. Identify the leading coefficient.

10. $-y^3 + 3y - 3y^2 + 2$

11. $11t + 2t^2 - 3 + t^5$

12. $2 + r - r^3$

13. $\frac{1}{2}x - 3x^4 + 7$

14. $-9b^2 + 10b - b^6$

Examples 3–5

Find each sum or difference.

22. $(2h^2 - 5h) + (7h - 3h^2)$

23. $(5f + g - 2) + (-2f + 3)$

24. $(6k^2 + 2k + 9) + (4k^2 - 5k)$

25. $(2c^2 + 6c + 4) + (5c^2 - 7)$

26. $(2x + 3x^2) - (7 - 8x^2)$

◆ LESSON 11-3 · Multiplying Polynomials

Examples 1–3

Find each product.

1. $(3c - 5)(c + 3)$

2. $(g + 10)(2g - 5)$

3. $(6a + 5)(5a + 3)$

4. $(4x + 1)(6x + 3)$

5. $(5y - 4)(3y - 1)$

6. $(6d - 5)(4d - 7)$

◆ LESSON 11-6 · Factoring Quadratic Trinomials

Examples 1–4

Factor each polynomial, if possible. If the polynomial cannot be factored using integers, write *prime*.

1. $x^2 + 17x + 42$

2. $y^2 - 17y + 72$

3. $a^2 + 8a - 48$

4. $n^2 - 2n - 35$

5. $44 + 15h + h^2$

6. $40 - 22x + x^2$

◆ LESSON 11-7 · Factoring Special Products

Examples 4 and 5

Determine whether each trinomial is a perfect square trinomial. Write *yes* or *no*. If so, factor it.

18. $4x^2 - 42x + 110$

19. $16x^2 - 56x + 49$

20. $81x^2 - 90x + 25$

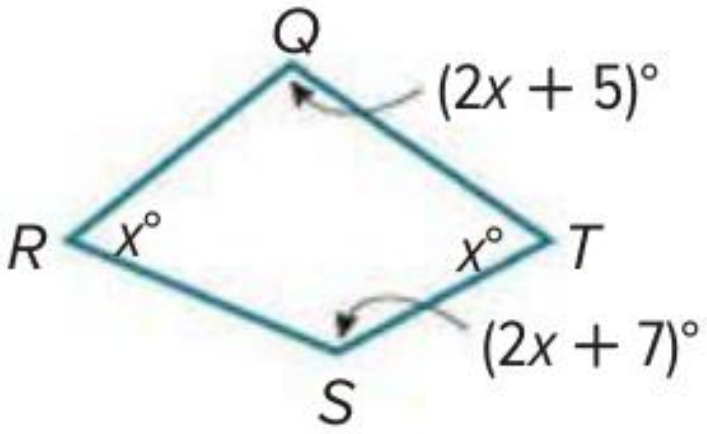
21. $x^2 + 26x + 168$

◆ LESSON 2-1 · Angles of Polygons

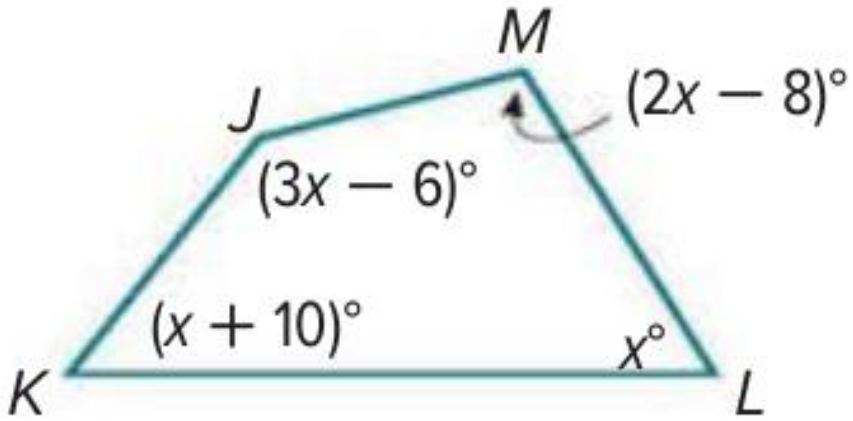
Example 1

Find the measure of each interior angle.

1.

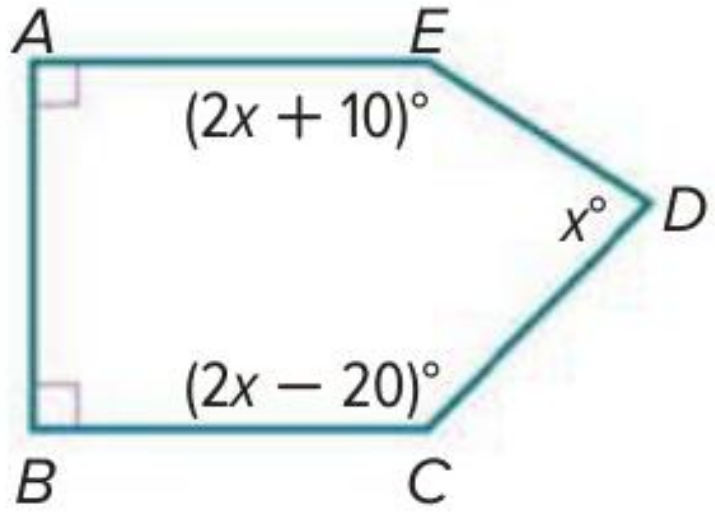


2.

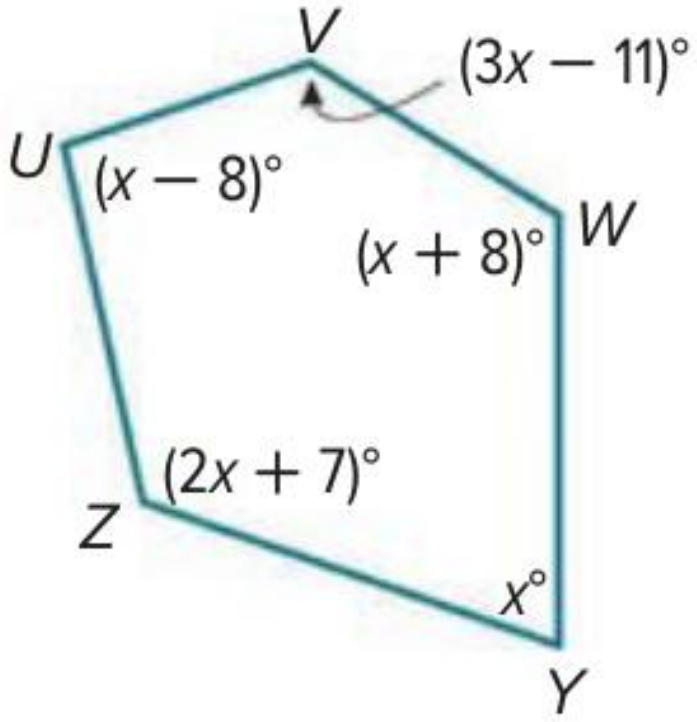


◆ LESSON 2-1 · Angles of Polygons

3.



4.

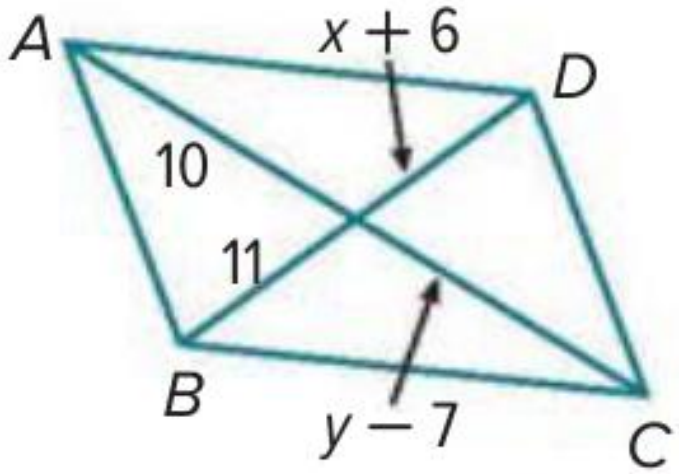


◆ LESSON 2-2 · Parallelograms

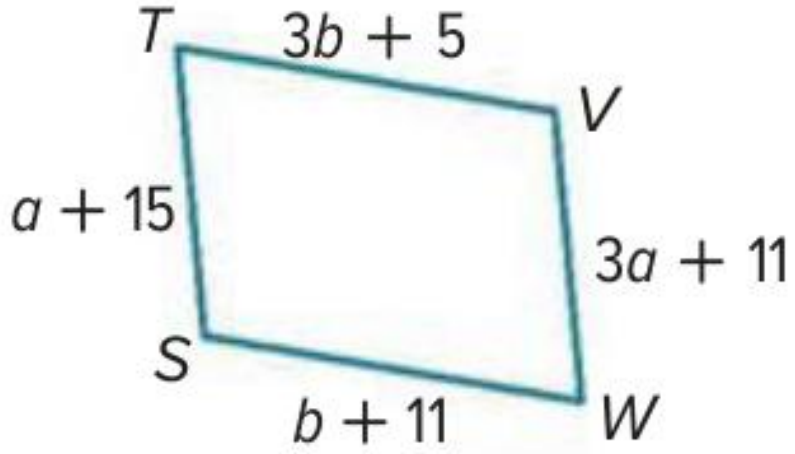
Example 3

Find the value of each variable in each parallelogram.

9.

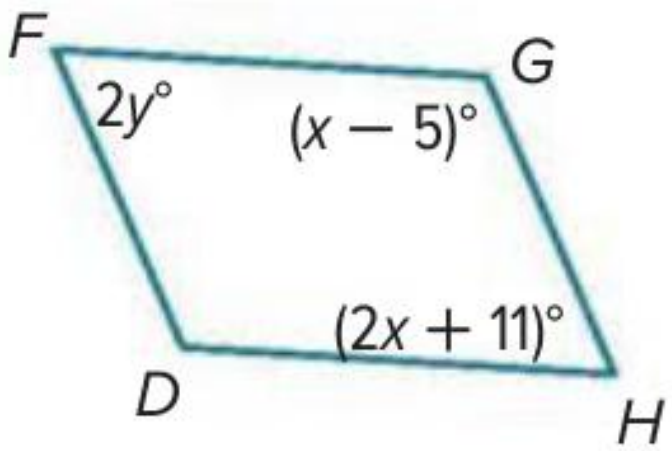


10.

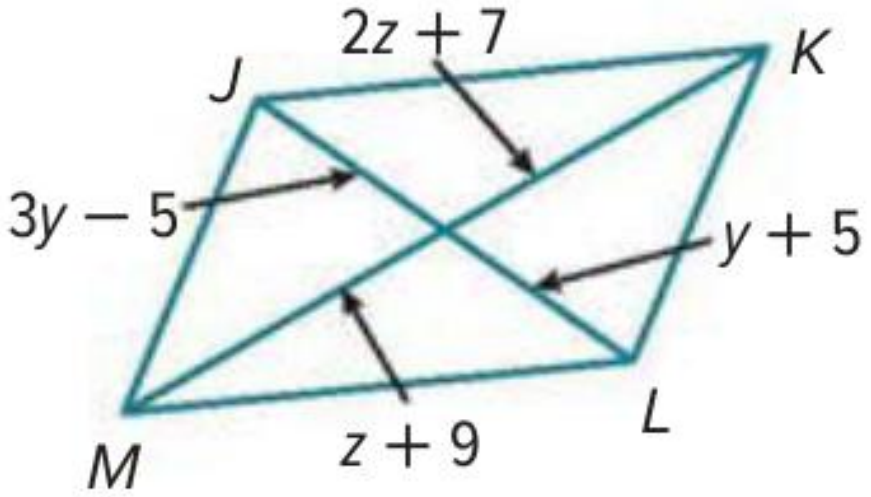


◆ LESSON 2-2 · Parallelograms

11.



12.



Learn Tests for Parallelograms

If a quadrilateral has each pair of opposite sides parallel, it is a parallelogram by definition. This is not the only test, however, that can be used to determine if a quadrilateral is a parallelogram.

Theorems: Conditions for Parallelograms

Theorem 2.9

If both pairs of opposite sides of a quadrilateral are congruent, then the quadrilateral is a parallelogram.

Theorem 2.10

If both pairs of opposite angles of a quadrilateral are congruent, then the quadrilateral is a parallelogram.

Theorem 2.11

If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.

Theorem 2.12

If one pair of opposite sides of a quadrilateral is both parallel and congruent, then the quadrilateral is a parallelogram.

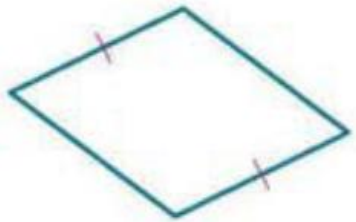
◆ LESSON 2-3 · Tests for Parallelograms

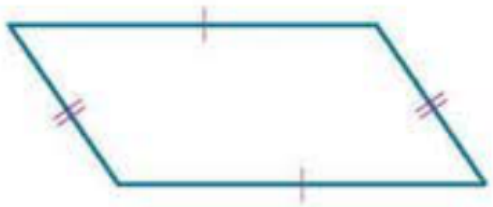
Example 1

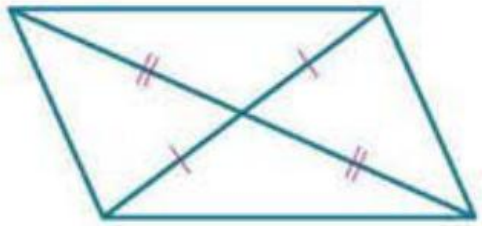
Determine whether each quadrilateral is a parallelogram. Justify your answer.

1. 

2. 

3. 

4. 

5. 

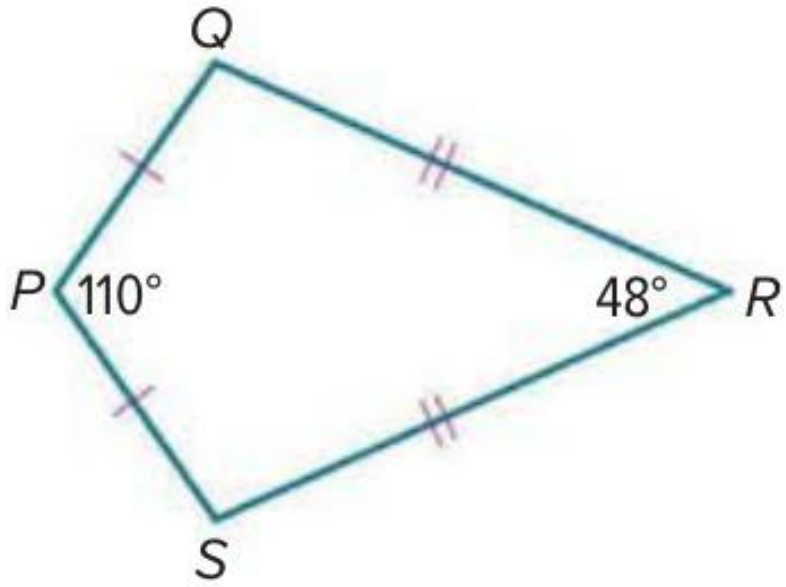
6. 

◆ LESSON 2-6 · Trapezoids and Kites

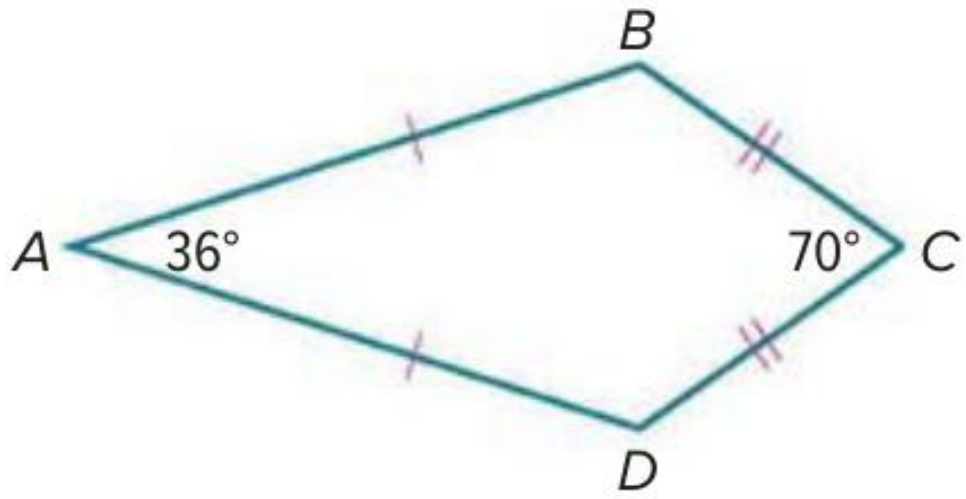
Example 6

Find each measure in the kites.

13. $m\angle Q$



14. $m\angle D$

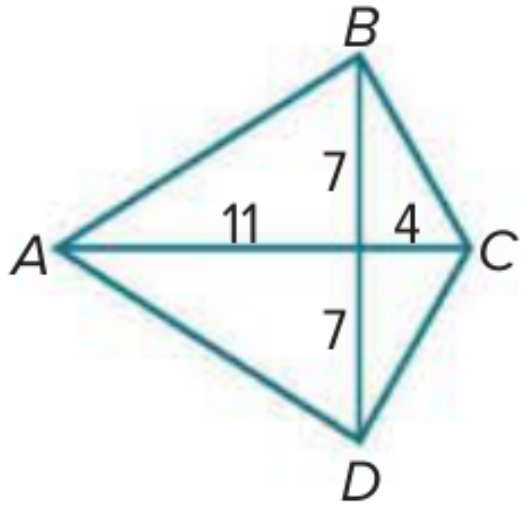


Example 7

15. REASONING Quadrilateral *ABCD* is a kite.

a. Find *BC*. Write your answer in simplest radical form.

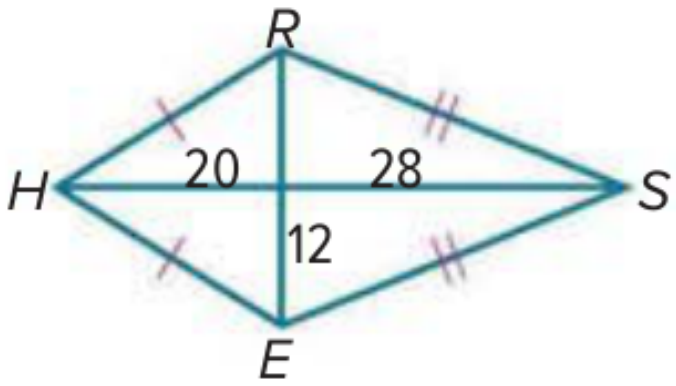
b. Find the perimeter of kite *ABCD*. Round your answer to the nearest tenth, if necessary.



16. REASONING Quadrilateral *HRSE* is a kite.

a. Find *RH*. Write your answer in simplest radical form.

b. Find the perimeter of kite *HRSE*. Round your answer to the nearest tenth, if necessary.



Mixed Exercises

Graph the image of each polygon with the given vertices after a dilation centered at the origin with the given scale factor.

13. $F(-10, 4)$, $G(-4, 4)$, $H(-4, -8)$, $k = 0.25$

14. $X(2, -1)$, $Y(-6, 4)$, $Z(-2, -5)$, $k = \frac{5}{4}$

15. $M(4, 6)$, $N(-6, 2)$, $P(0, -8)$, $k = \frac{3}{4}$

16. $R(-2, 6)$, $S(0, -1)$, $T(-5, 3)$, $k = 1.5$

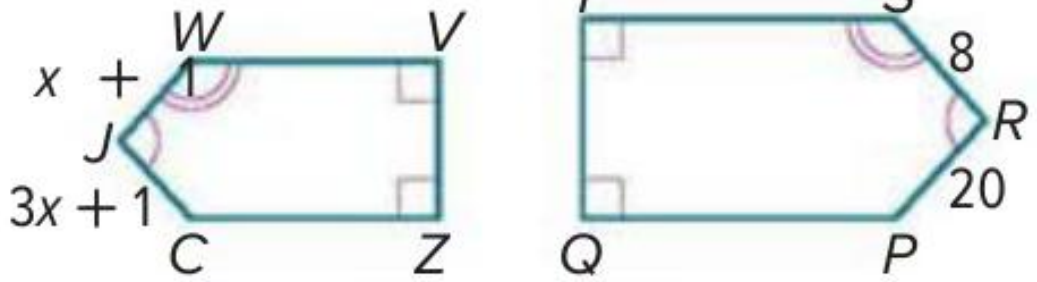
20. $\triangle ABC$ has vertices $A(2, 2)$, $B(3, 4)$, and $C(5, 2)$. What are the coordinates of point C of the image of the triangle after a dilation centered at the origin with a scale factor of 2.5?

◆ LESSON 3-2 · Similar Polygons

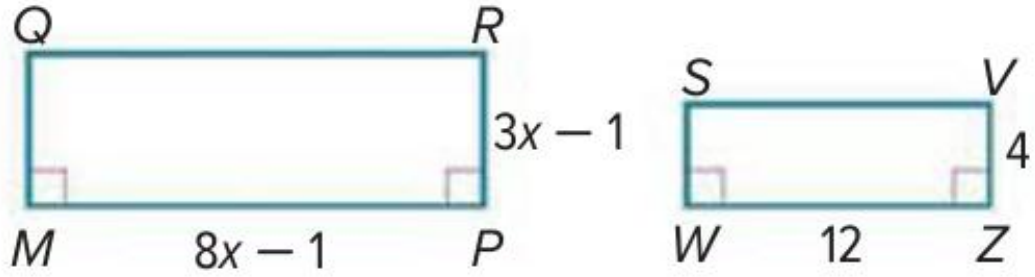
Example 3

Each pair of polygons is similar. Find the value of *x*.

11.



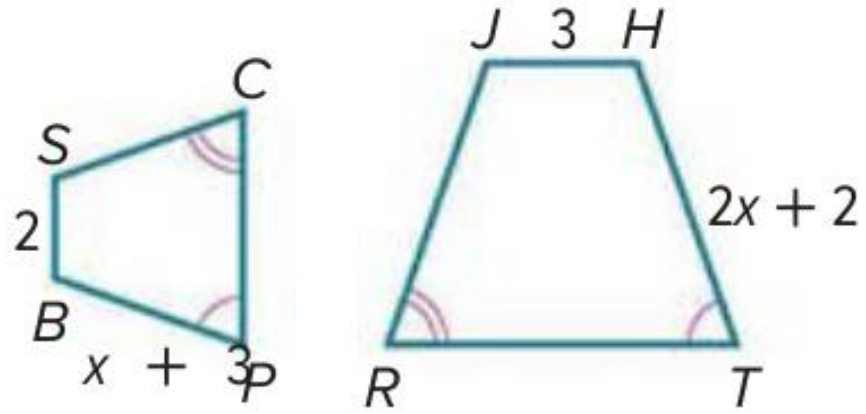
12.



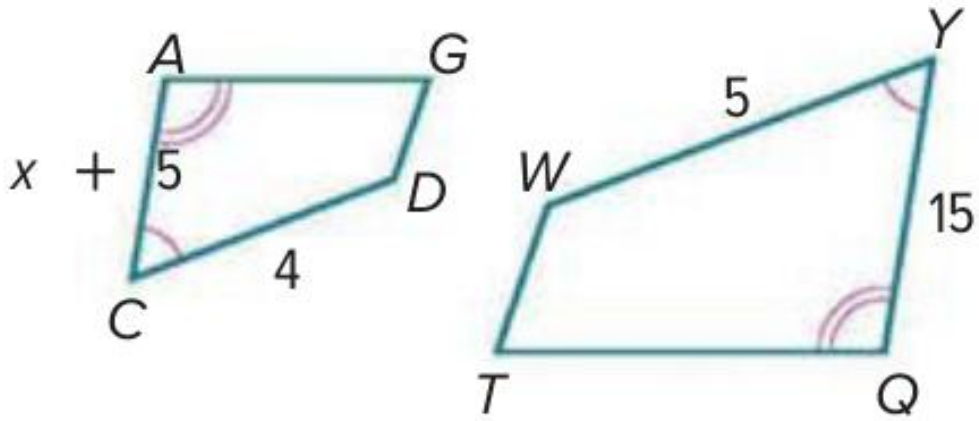
◆ LESSON 3-2 · Similar Polygons

PART 2 / 2

13.



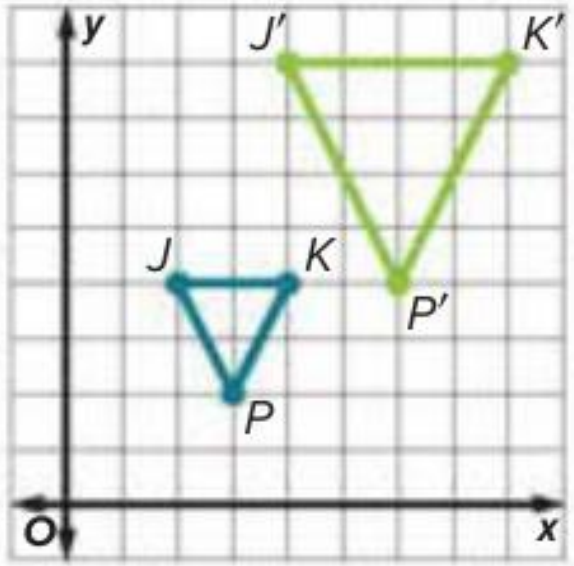
14.



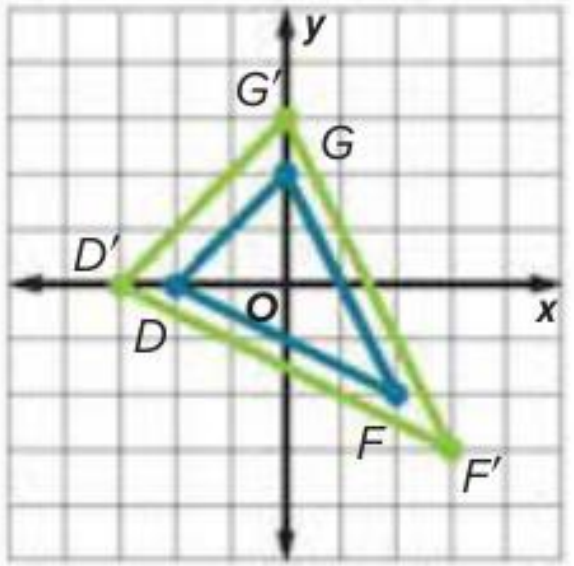
Example 4

Find the scale factor of the dilation.

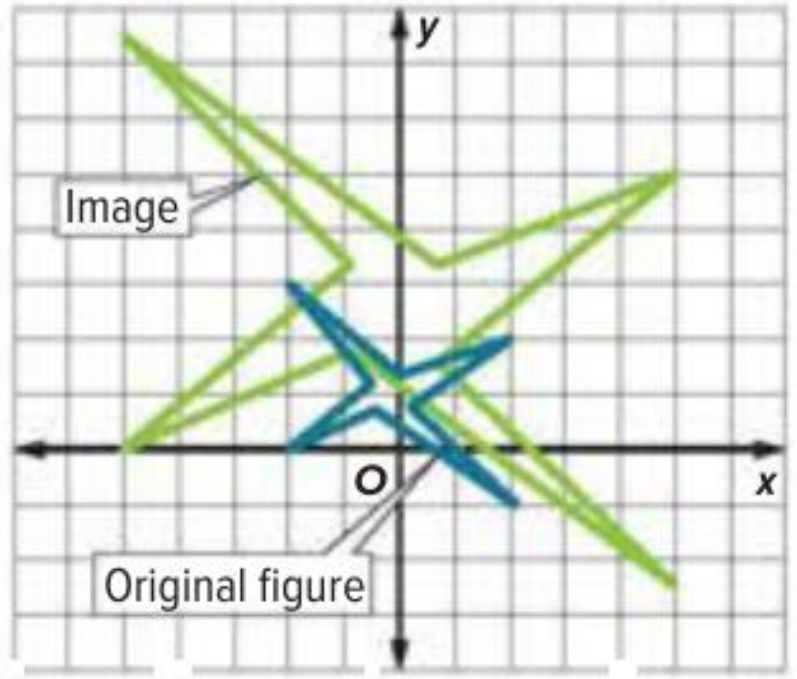
10. $\triangle J'K'P'$ is the image of $\triangle JKP$.



11. $\triangle D'F'G'$ is the image of $\triangle DFG$.

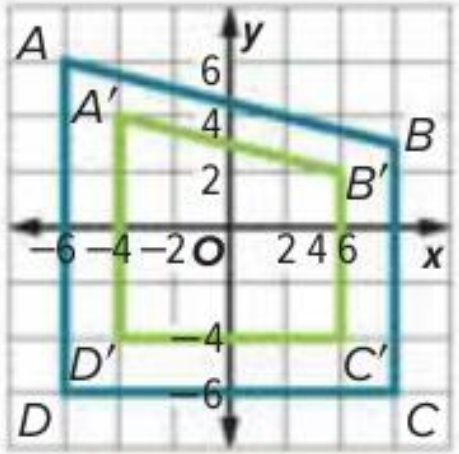


12. Tyrone drew a logo and a dilation of the same logo on the coordinate plane. What is the scale factor of the dilation?

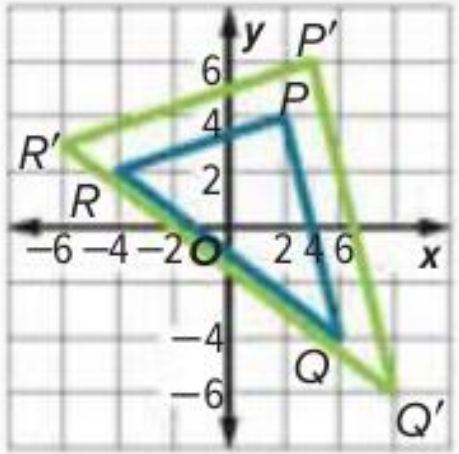


Find the scale factor of the dilation.

17. $A'B'C'D'$ is the image of $ABCD$.



18. $\triangle P'Q'R'$ is the image of $\triangle PQR$.



Identify similar triangles using the AA Similarity Postulate

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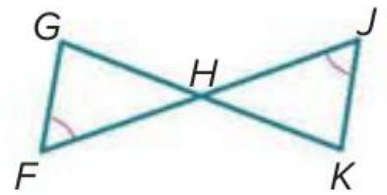
MCQ Exercises 1-8 · P 133

◆ LESSON 3-3 · Similar Triangles: AA Similarity

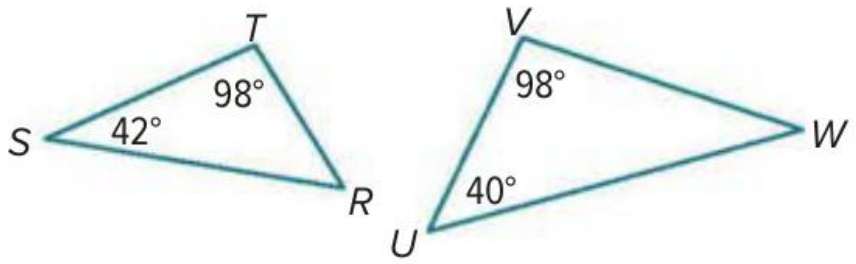
Example 1

Determine whether each pair of triangles is similar. Explain your reasoning.

1.



2.



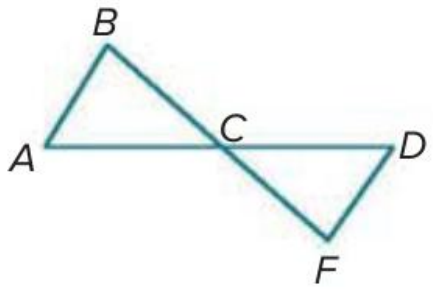
Identify similar triangles using the AA Similarity Postulate

تحديد المثلثات المتشابهة باستخدام مسلّمة التشابه AA

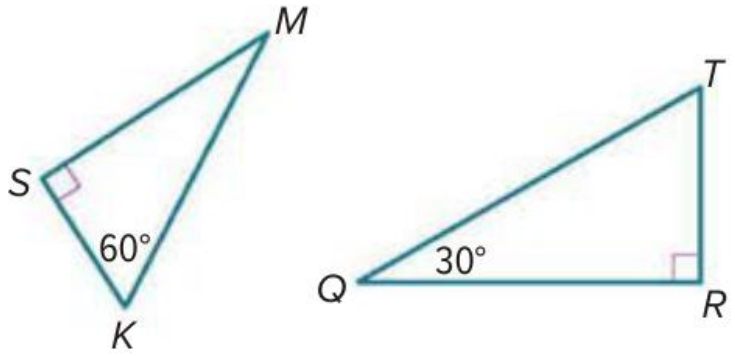
MCQ Exercises 1-8 · P 133

◆ LESSON 3-3 · Similar Triangles: AA Similarity

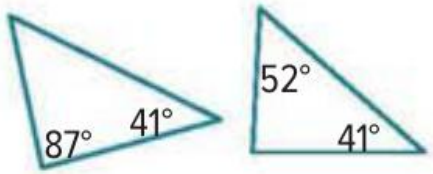
3.



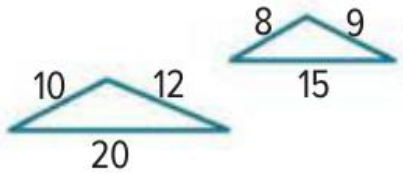
4.



5.



6.



◆ LESSON 3-3 · Similar Triangles: AA Similarity

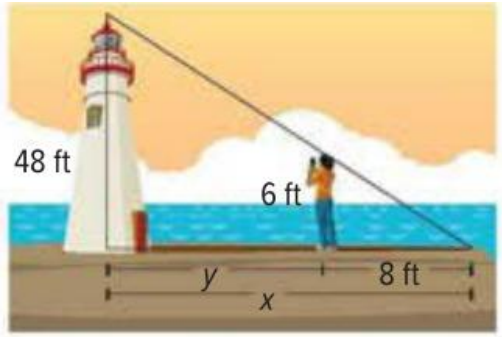
Example 2

7. CELL TOWERS A cell phone tower casts a shadow that is 100 feet long. At the same time, Lia stands near the tower and casts a shadow that is 3 feet 4 inches long. If Lia is 4 feet 6 inches tall, how tall is the cell phone tower?

8. LIGHTHOUSE Maya wants to know how far she is standing from a lighthouse. The end of Maya's shadow coincides with the end of the lighthouse's shadow.

a. What is the distance from the lighthouse to the end of the lighthouse's shadow, x?

b. What is the distance from Maya to the lighthouse, y?

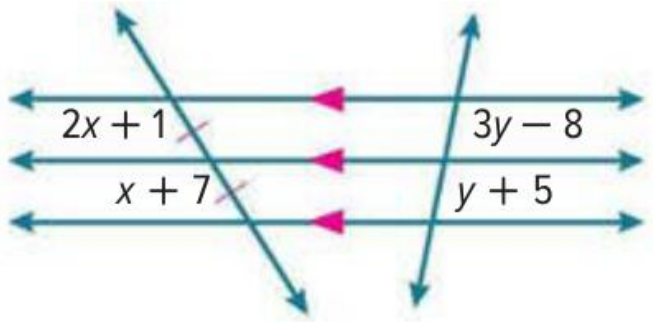


◆ LESSON 3-5 · Triangle Proportionality

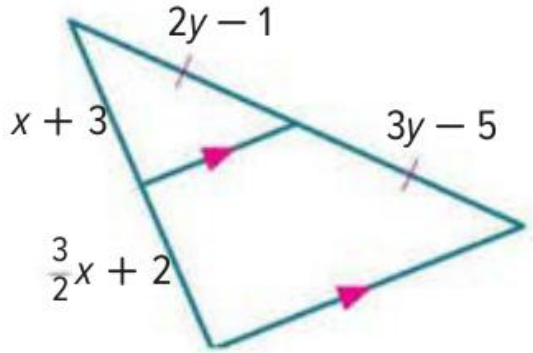
Example 5

Find the values of *x* and *y*.

11.



12.

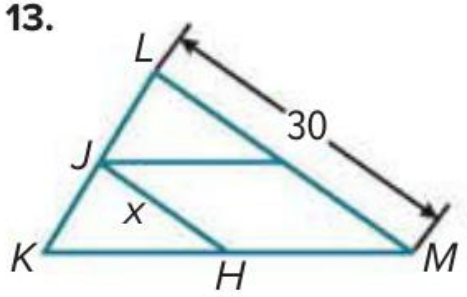


◆ LESSON 3-5 · Triangle Proportionality

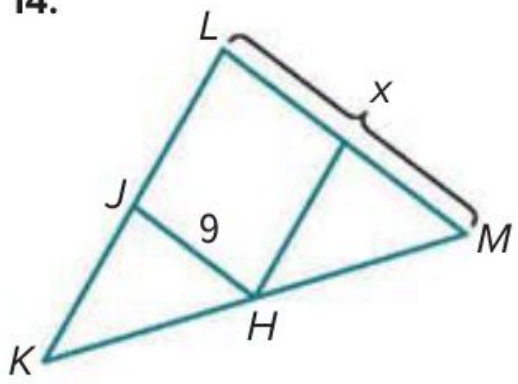
Mixed Exercises

$\overline{JH}$ is a midsegment of $\triangle KLM$. Find the value of x.

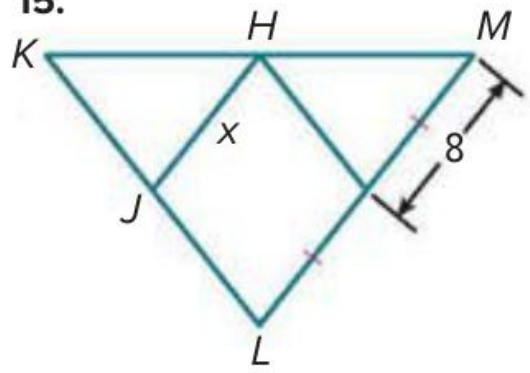
13.



14.



15.



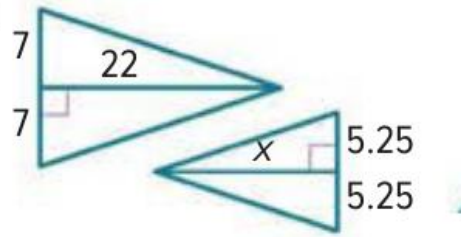
16. In $\triangle ABC$, $\overline{DE}$ is parallel to $\overline{AC}$ and $DE = \frac{1}{2}AC$ if $\overline{DE}$ is a midsegment of $\triangle ABC$.

◆ LESSON 3-6 · Parts of Similar Triangles

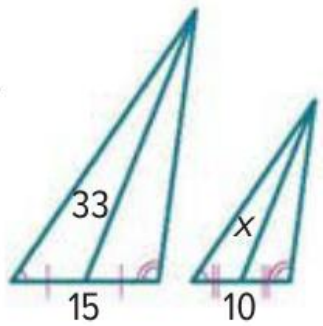
Example 1

Each pair of triangles is similar. Find the value of *x*.

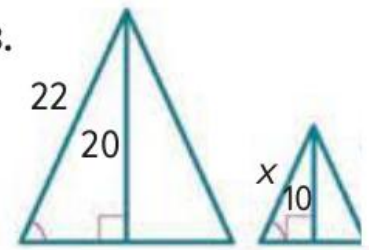
1.



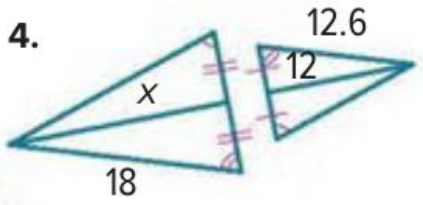
2.



3.



4.



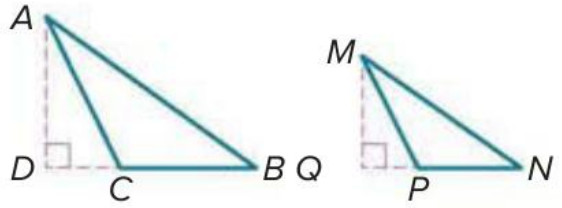
◆ LESSON 3-6 · Parts of Similar Triangles

Example 2

5. If $\triangle RST \sim \triangle EFG$, $S\overline{H}$ is an altitude of $\triangle RST$, $\overline{FJ}$ is an altitude of $\triangle EFG$, $ST = 6$, $SH = 5$, and $FJ = 7$, find FG.

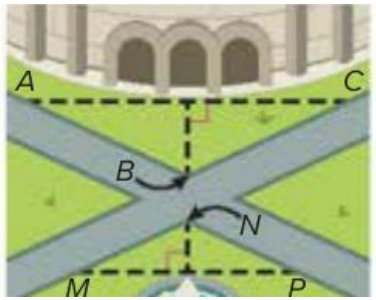


6. If $\triangle ABC \sim \triangle MNP$, $\overline{AD}$ is an altitude of $\triangle ABC$, $\overline{MQ}$ is an altitude of $\triangle MNP$, $AB = 24$, $AD = 14$, and $MQ = 10.5$, find MN.



◆ LESSON 3-6 · Parts of Similar Triangles

7. SCENERY The intersection of the two paths shown forms two similar triangles. If *AC* is 50 yards, *MP* is 35 yards, and the fountain is 5 yards from the intersection, about how far from the intersection is the stadium entrance? Round to the nearest yard.

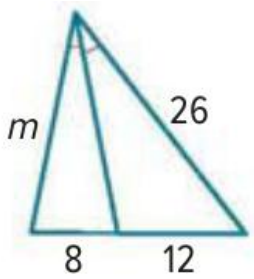


◆ LESSON 3-6 · Parts of Similar Triangles

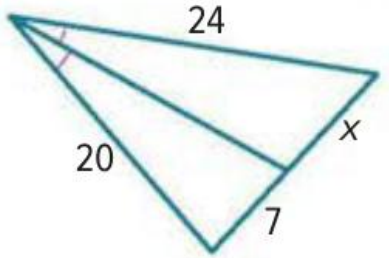
Example 3

Find the value of each variable to the nearest tenth.

8.



9.

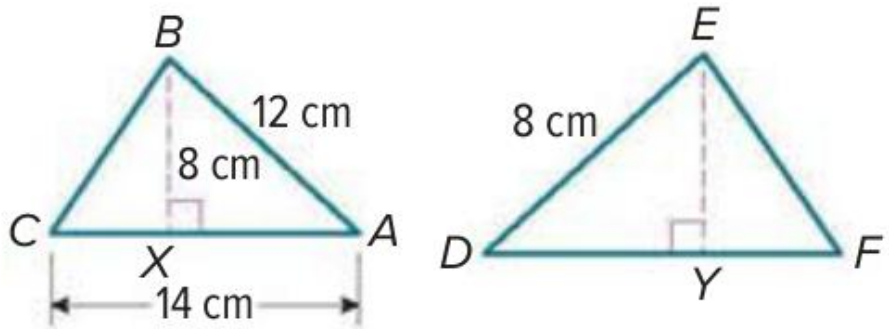


◆ LESSON 3-6 · Parts of Similar Triangles

Mixed Exercises

For Exercises 10 and 11, $\triangle ABC \sim \triangle DEF$.

10. Find the length of $\overline{XC}$ to the nearest tenth.



11. Find the length of $\overline{EY}$ to the nearest tenth.

◆ LESSON 4-1 · Geometric Mean

Example 1

Find the geometric mean between each pair of numbers.

1. 4 and 6

2. $\frac{1}{2}$ and 2

3. 4 and 25

4. 12 and 20

5. 17 and 3

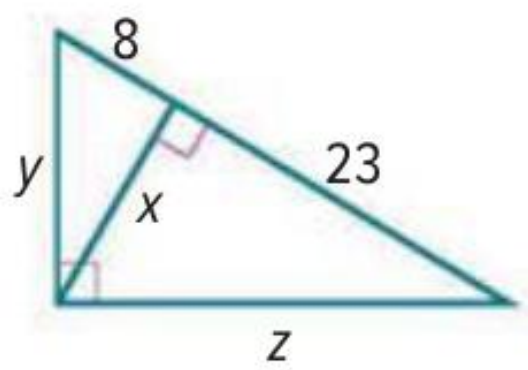
6. 3 and 24

◆ LESSON 4-1 · Geometric Mean

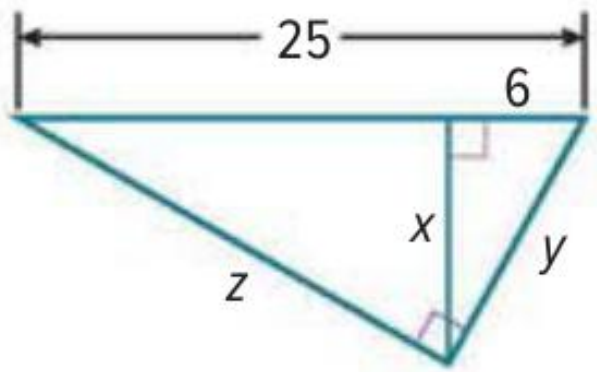
Example 3

Find the values of *x*, *y*, and *z*.

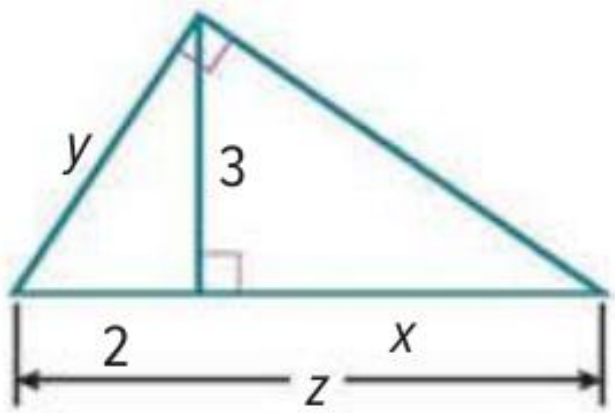
11.



12.

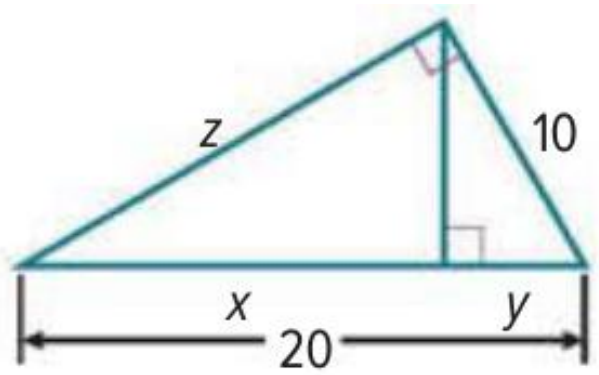


13.



◆ LESSON 4-1 · Geometric Mean

14.

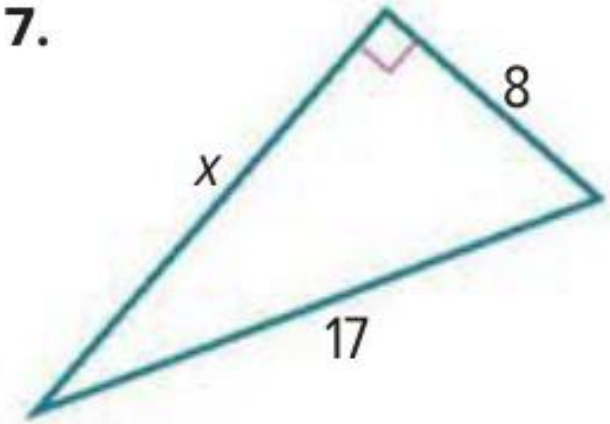


◆ LESSON 4-2 · Pythagorean Theorem and Its Converse

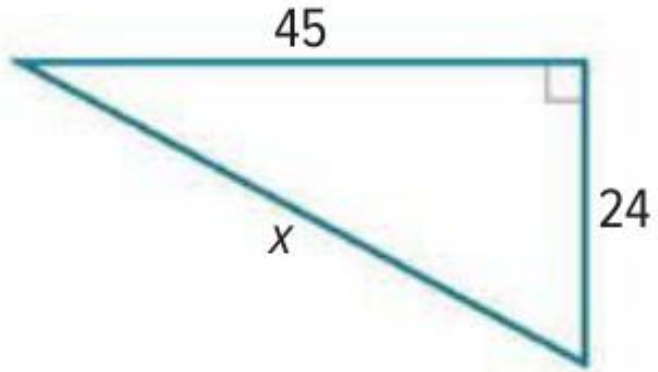
Example 2

Use a Pythagorean Triple to find the value of *x*.

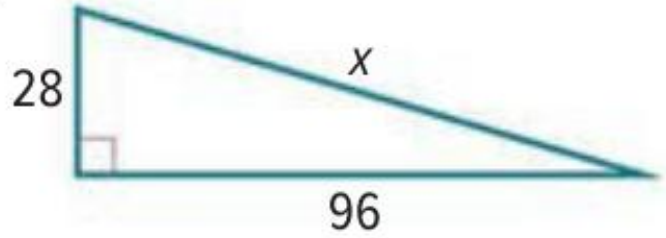
7.



8.

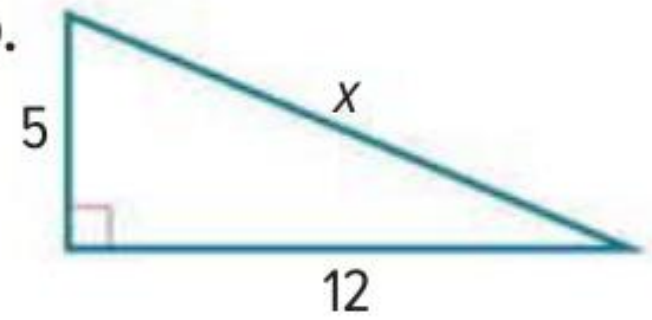


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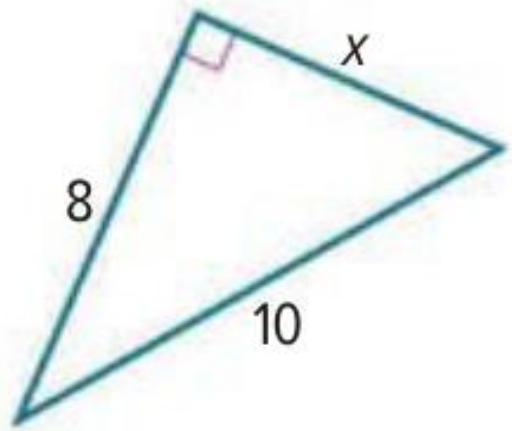


◆ LESSON 4-2 · Pythagorean Theorem and Its Converse

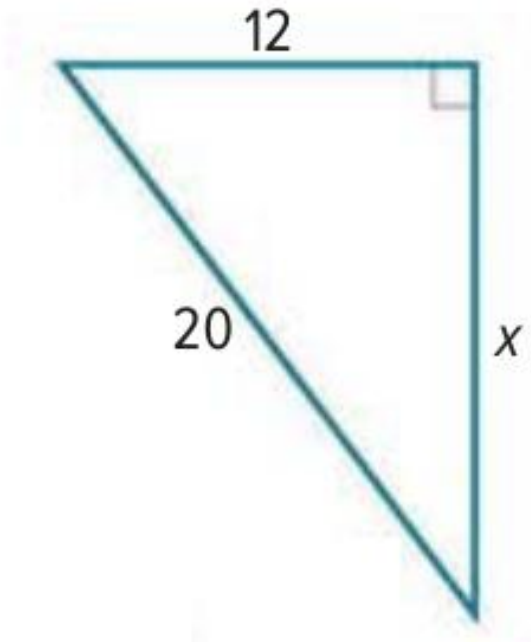
10.



11.



12.



◆ LESSON 4-2 · Pythagorean Theorem and Its Converse

Determine whether each set of measures can be the measures of the sides of a triangle. If so, classify the triangle as *acute*, *obtuse*, or *right*. Justify your answer.

20. $\sqrt{5}, \sqrt{12}, \sqrt{13}$

21. $2, \sqrt{8}, \sqrt{12}$

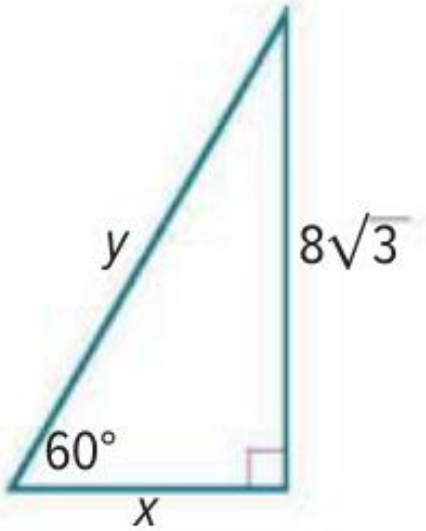
22. 9, 40, 41

◆ LESSON 4-4 · Special Right Triangles

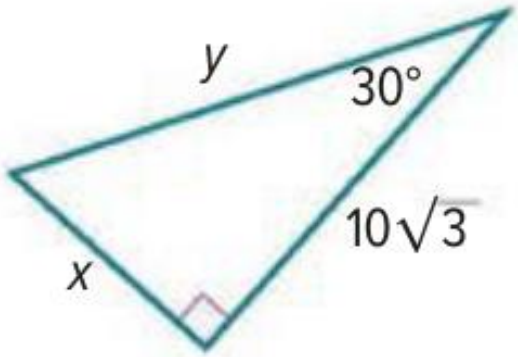
Example 4

Find the values of *x* and *y*.

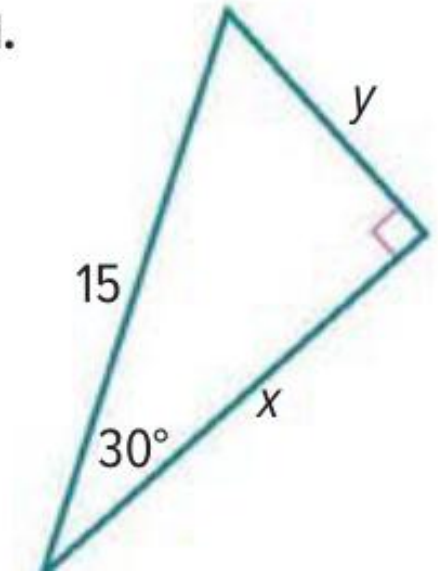
19.



20.

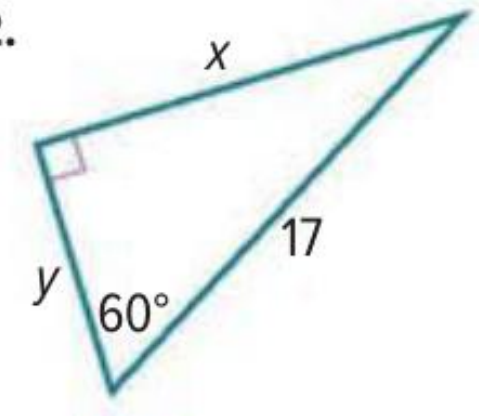


21.

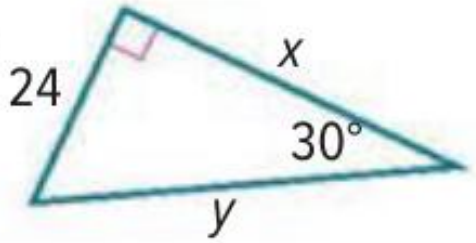


◆ LESSON 4-4 · Special Right Triangles

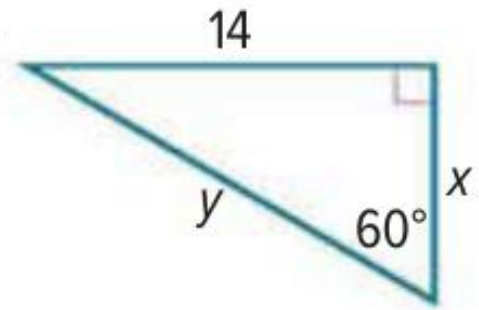
22.



23.



24.



Example 5

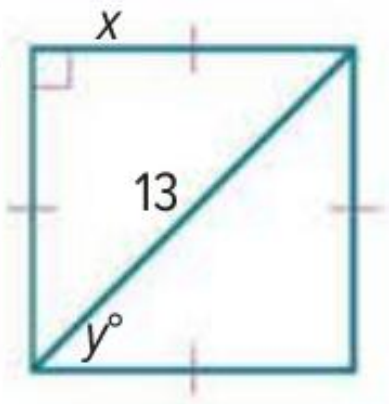
25. ESCALATORS A 40-foot-long escalator rises from the first floor to the second floor of a shopping mall. The escalator makes a 30° angle with the horizontal. How high above the first floor is the second floor?



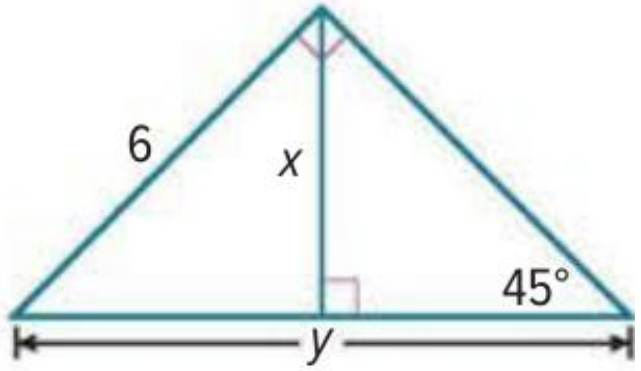
◆ LESSON 4-4 · Special Right Triangles

Find the values of *x* and *y*.

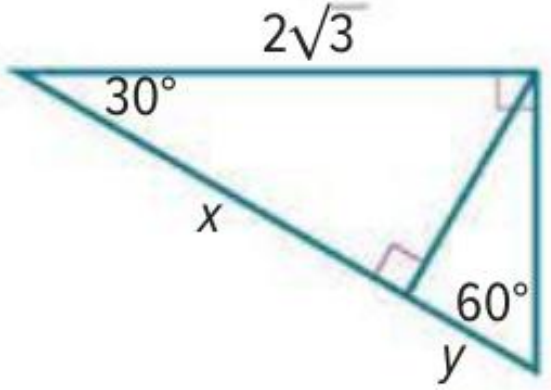
33.



34.

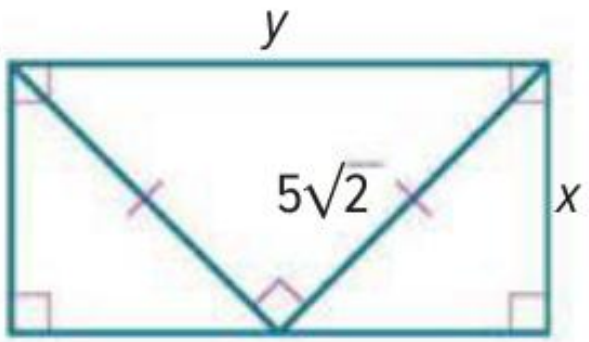


35.

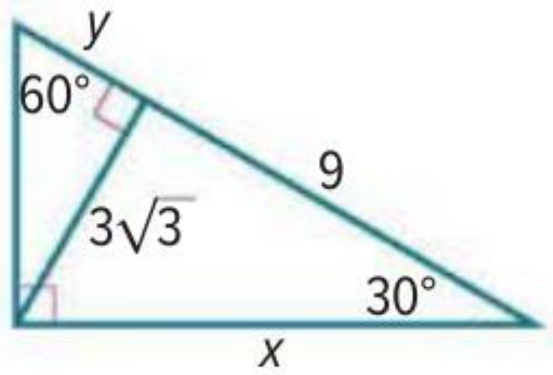


◆ LESSON 4-4 · Special Right Triangles

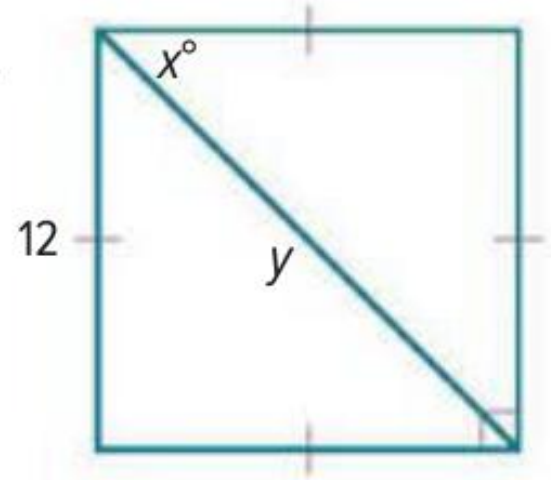
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37.



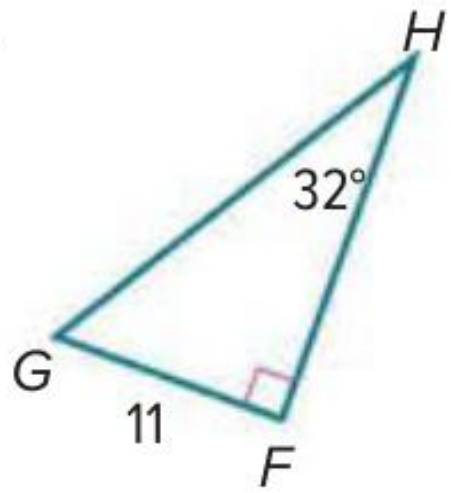
38.



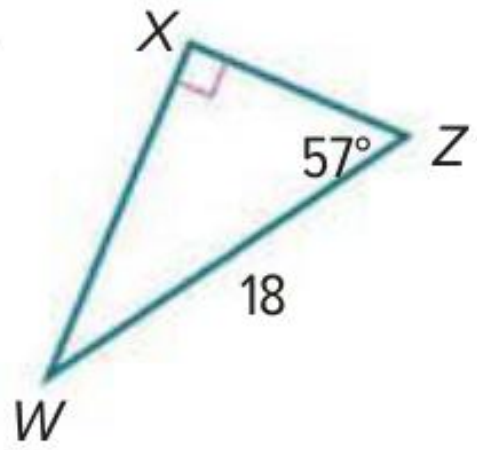
Example 5

Solve each right triangle. Round side measures to the nearest tenth and angle measures to the nearest degree.

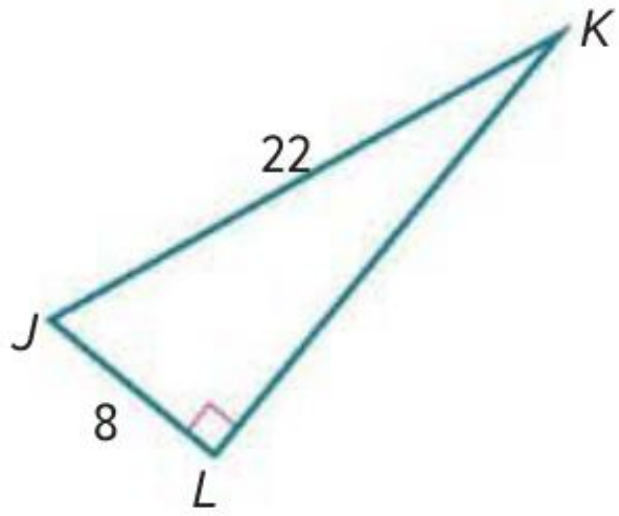
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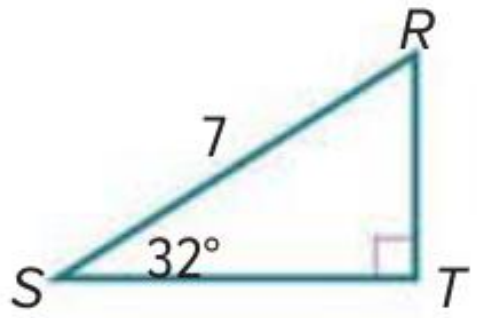


24.

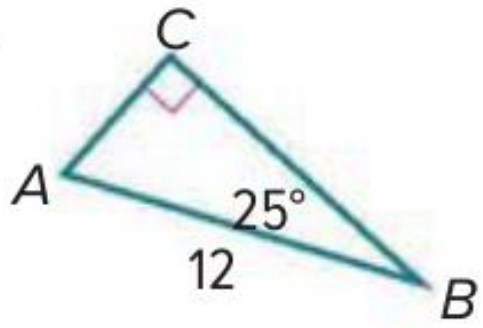


◆ LESSON 4-5 · Trigonometry

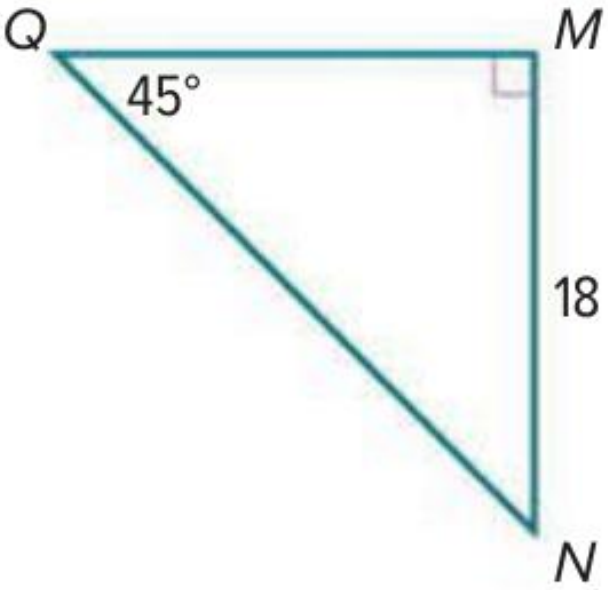
25.



26.

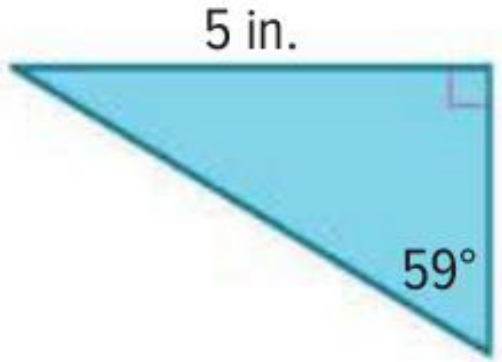


27.

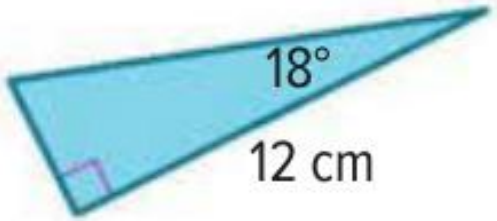


◆ LESSON 4-5 · Trigonometry

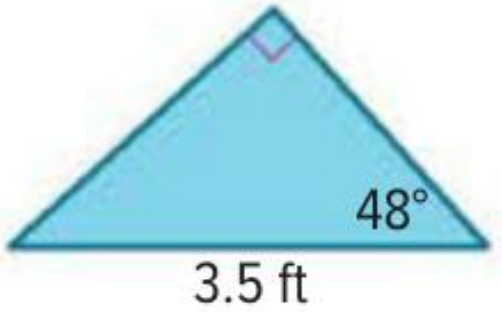
30.



31.



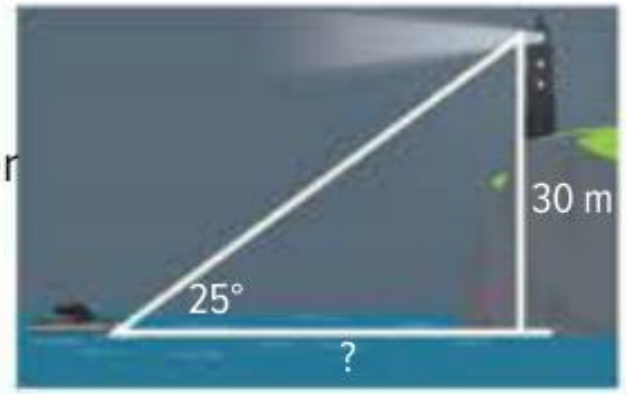
32.



◆ LESSON 4-6 · Applying Trigonometry

Example 1

1. LIGHTHOUSES Sailors on a ship at sea spot the light from a lighthouse at an angle of elevation of 25°. The light of the lighthouse is 30 meters above sea level. How far from the shore is the ship? Round your answer to the nearest meter.



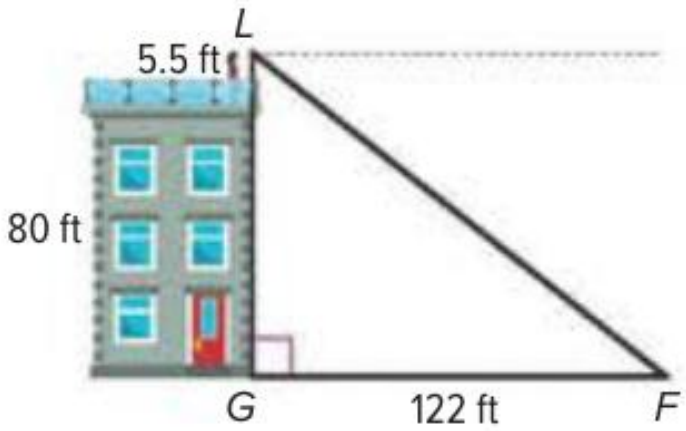
2. **WATER TOWERS** A student can see a water tower from the edge of a soccer field at San Lobos High School. The edge of the field is about 110 feet from the water tower, and the water tower stands at a height of 32.5 feet. What is the angle of elevation if the eye level of the student viewing the tower is 6 feet above the ground? Round your answer to the nearest tenth.

3. **CONSTRUCTION** A roofer props a ladder against a wall so the top of the ladder just reaches a 30-foot roof that needs repair. If the angle of elevation from the bottom of the ladder to the roof is 55°, how far is the ladder from the base of the wall? Round your answer to the nearest foot.

4. MOUNTAIN BIKING On a mountain bike trip along the Gemini Bridges Trail in Moab, Utah, Nabuko stopped on the canyon floor to get a good view of the twin sandstone bridges. Nabuko is standing about 60 meters from the base of the canyon cliff, and the natural arch bridges are about 100 meters up the canyon wall. If her line of sight is 5 meters above the ground, what is the angle of elevation to the top of the bridges? Round to the nearest tenth of a degree.

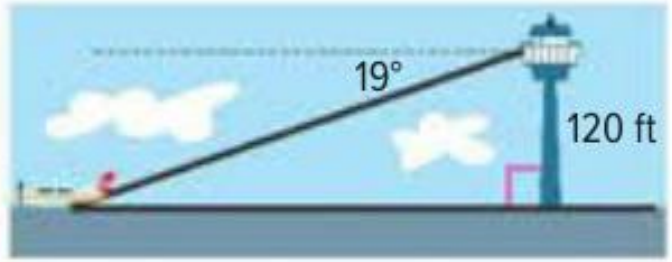
Example 2

5. ROOFTOP Lucia is 5.5 feet tall. She is standing on the roof of a building that is 80 feet tall. She spots a fountain at ground level that she knows to be 122 feet away from the base of the building. What is the measure of the angle of depression formed by Lucia's horizontal line of sight and her line of sight to the fountain? Round your answer to the nearest degree.



◆ LESSON 4-6 · Applying Trigonometry

6. AIR TRAFFIC From the top of the 120-foot-high tower, an air traffic controller observes an airplane on the runway at an angle of depression of 19°. How far from the base of the tower is the airplane? Round to the nearest tenth of a foot.



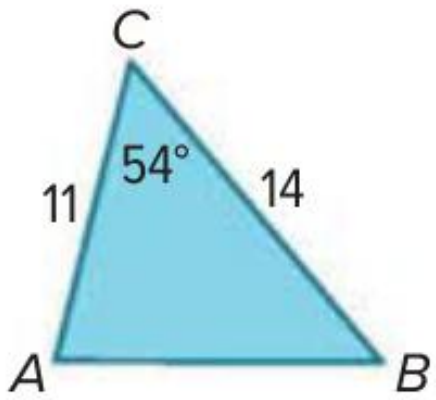
7. AVIATION Due to a storm, a pilot flying at an altitude of 528 feet has to land. If he has a horizontal distance of 2000 feet until he reaches the landing strip, what angle of depression should he use to land? Round to the nearest tenth of a degree.

◆ LESSON 4-6 · Applying Trigonometry

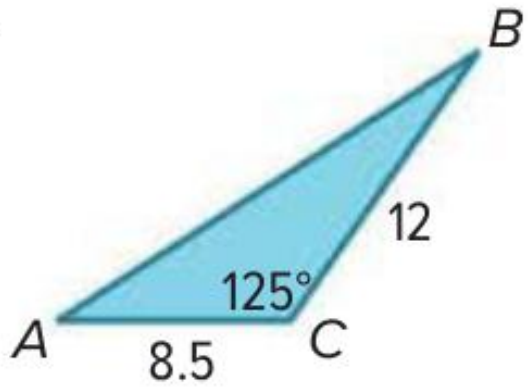
Example 4

Use trigonometry to find the area of $\triangle ABC$ to the nearest tenth.

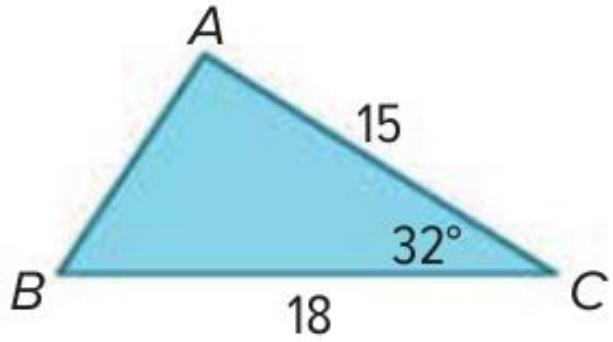
13.



14.



15.

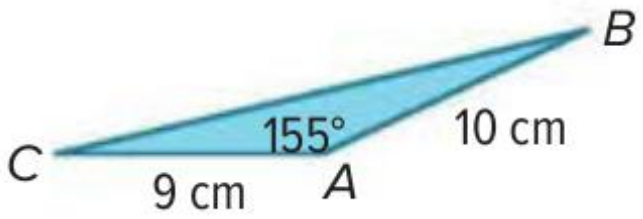


Find the area of a triangle using two sides and an included angle

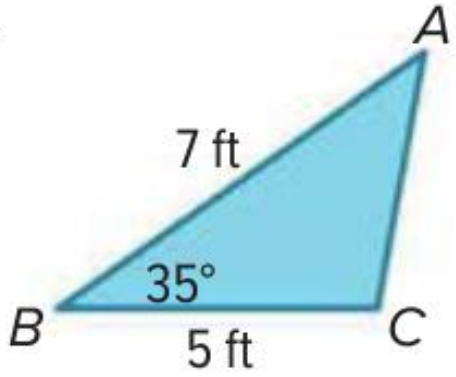
إيجاد مساحة مثلث بمعلومية ضلعين والزاوية المحصورة بينهما

MCQ Exercises 13-21 · P 200–201

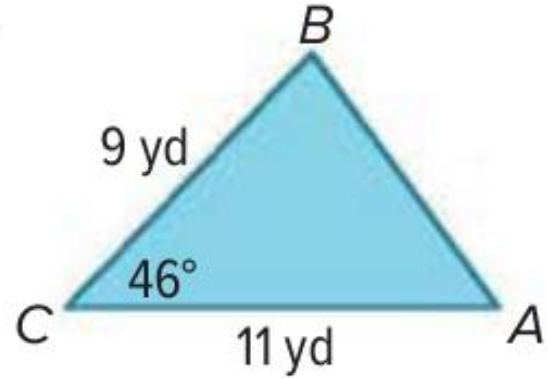
16.



17.



18.

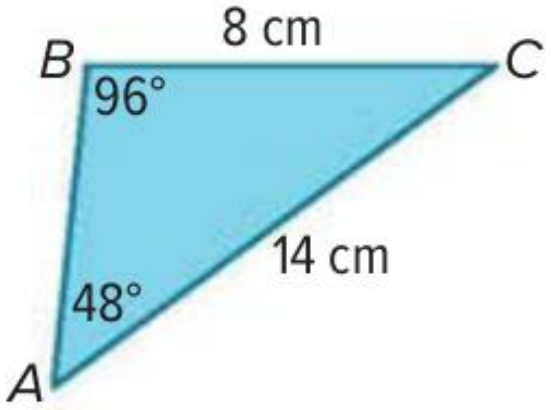


◆ LESSON 4-6 · Applying Trigonometry

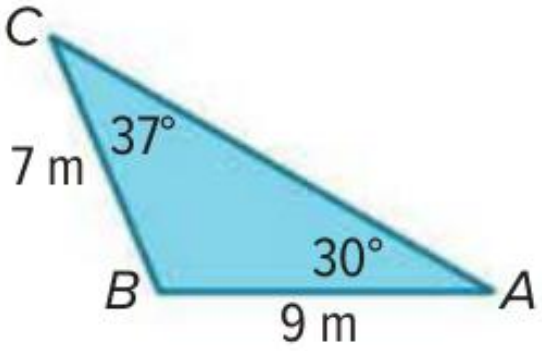
Example 5

Use trigonometry to find the area of $\triangle ABC$ to the nearest tenth.

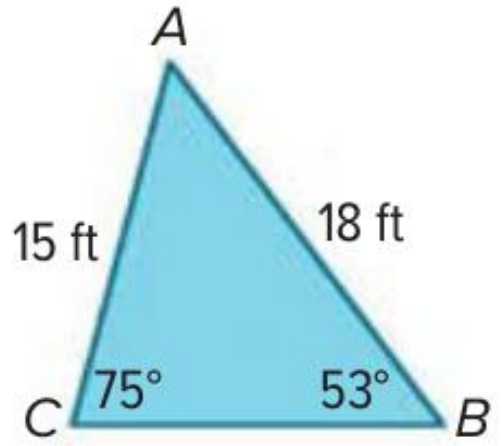
19.



20.



21.



Mixed Exercises

Factor each polynomial, if possible. If the polynomial cannot be factored using integers, write *prime*.

35. $8x^2 - 72p^2$

36. $20q^2 - 5r^2$

37. $32a^2 - 50b^2$

38. $16b^2 - 100$

39. $49x^2 - 64y^2$

40. $3n^4 - 42n^3 + 147n^2$

41. $8m^3 - 24m^2 + 18m$

Determine whether each trinomial is a perfect square trinomial. Write *yes* or *no*. If so, factor it.

51. $9z^2 - 6z + 1$

52. $36x^2 - 60x + 25$

53. $49r^2 + 14r + 4$

54. $a^2 + 14a + 49$

55. $t^2 - 18t + 81$

56. $4c^2 + 2cd + d^2$

◆ LESSON 2-1 · Angles of Polygons

Example 3

The measure of an interior angle of a regular polygon is given. Find the number of sides in the polygon.

9. 144°

10. 156°

11. 160°

12. 108°

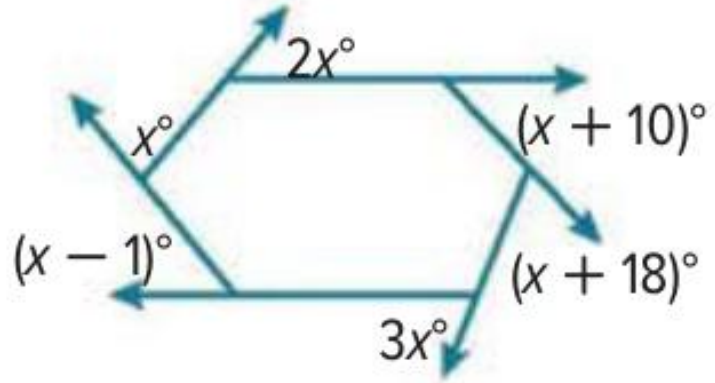
13. 120°

14. 150°

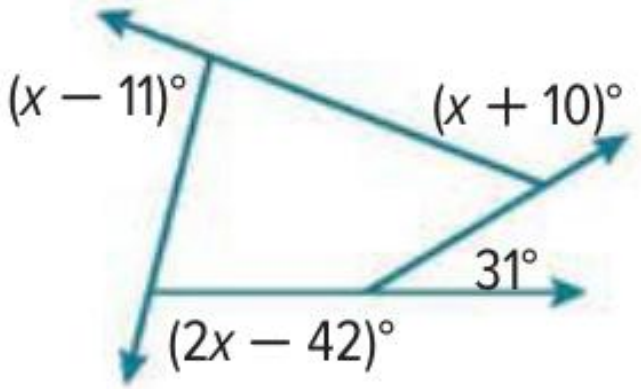
Example 4

Find the value of *x* in each diagram.

15.

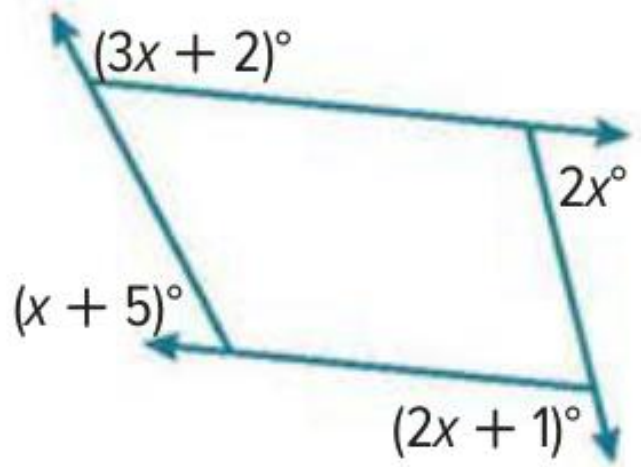


16.

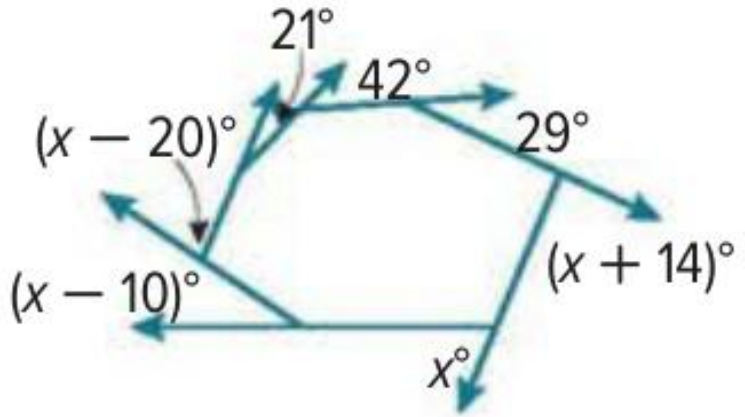


◆ LESSON 2-1 · Angles of Polygons

17.



18.

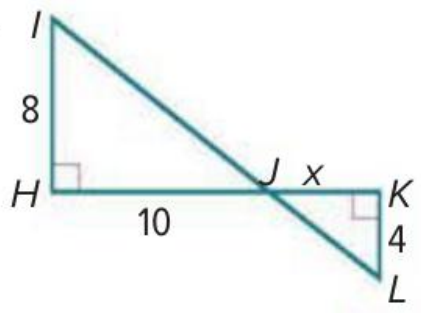


◆ LESSON 3-4 · Similar Triangles: SSS and SAS Similarity

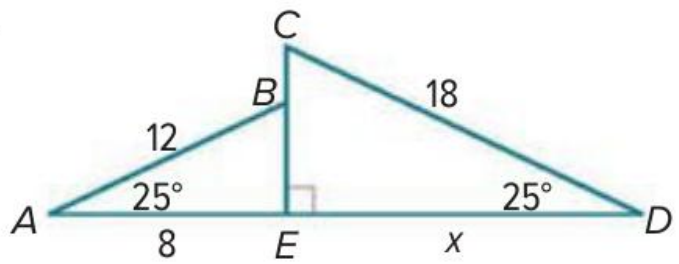
Example 2

Identify the similar triangles. Then find the value of *x*.

5.

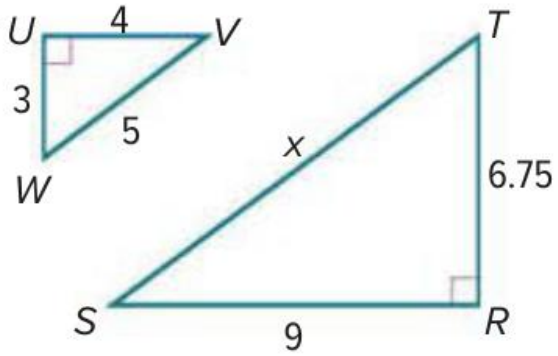


6.

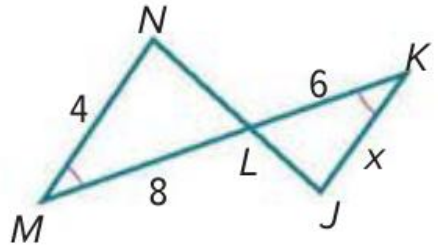


◆ LESSON 3-4 · Similar Triangles: SSS and SAS Similarity

7.



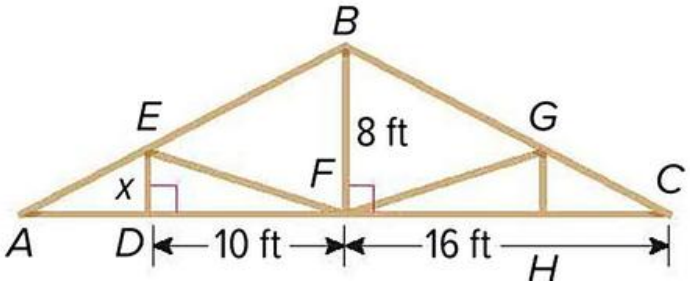
8.



◆ LESSON 3-4 · Similar Triangles: SSS and SAS Similarity

Example 3

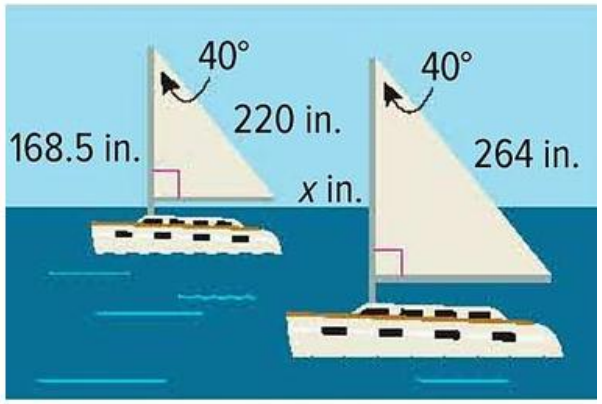
9. ROOFING The skeleton of a roof is shown. Find the value of x such that triangles DEF and FBC in the outline of the roof are similar



10. RADIO A radio tower casts an 8-foot-long shadow at the same time that a vertical yardstick casts a shadow one half inch long. If the triangles formed by the objects and their shadows are similar, how tall is the radio tower?

◆ LESSON 3-4 · Similar Triangles: SSS and SAS Similarity

11. SAILING The two sailboats shown are participating in a regatta. If the sails are similar, what is the value of x?



Example 3

Determine the coordinates of the midpoint *M* of the segment joining each pair of points.

13. $K(-2, -4, -4)$ and $L(4, 2, 0)$

14. $W(-1, -3, -6)$ and $Z(-1, 5, 10)$

15. $R(3, 3, 4)$ and $V(5, 4, 13)$

16. $A(4, 6, -8)$ and $B(0, 0, 0)$

17. $C(8, 7, 11)$ and $D(2, 1, 8)$

18. $T(-1, -7, 9)$ and $U(5, -1, -6)$

Mixed Exercises

REGULARITY **Determine the distance between each pair of points. Then determine the coordinates of the midpoint *M* of the segment joining the pair of points.**

19. $P(-5, -2, -1)$ and $Q(-1, 0, 3)$

20. $J(1, 1, 1)$ and $K(-1, -1, -1)$

21. $F\left(\frac{3}{5}, 0, \frac{4}{5}\right)$ and $G(0, 3, 0)$

22. $G(1, -1, 6)$ and $H\left(\frac{1}{5}, -\frac{2}{5}, 2\right)$

23. $B(\sqrt{3}, 2, 2\sqrt{2})$ and $C(-2\sqrt{3}, 4, 4\sqrt{2})$

24. $S(6\sqrt{3}, 4, 4\sqrt{2})$ and $T(4\sqrt{3}, 5, \sqrt{2})$

Examples 4–6

Determine whether each triangle has *no solution*, *one solution*, or *two solutions*. Then, solve the triangle. Round side lengths to the nearest tenth and angle measures to the nearest degree.

15. $m\angle A = 50°, a = 34, b = 40$

16. $m\angle A = 24°, a = 3, b = 8$

17. $m\angle A = 125°, a = 22, b = 15$

18. $m\angle A = 30°, a = 1, b = 4$

19. $m\angle A = 30°, a = 2, b = 4$

20. $m\angle A = 30°, a = 3, b = 4$

21. $m\angle A = 38^\circ, a = 10, b = 9$

22. $m\angle A = 78^\circ, a = 8, \quad b = 5$

23. $m\angle A = 133^\circ, a = 9, b = 7$

24. $m\angle A = 127^\circ, a = 2, b = 6$

25. $m\angle A = 109^\circ, a = 24, b = 13$

26. $m\angle A = 48^\circ, a = 11,$ and $b = 16$